

Clothes as Human Infrastructure: Materials, Functions, Culture, Sustainability, and Intelligent Futures

An Integrative Multidisciplinary Review

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Abstract

Clothing is often discussed as fashion, yet it is more accurately understood as a form of human infrastructure that connects the body to climate, culture, technology, labour and the biosphere. This integrative review develops a systems account of clothes across six linked domains: physiological protection and comfort; material and manufacturing science; identity and social meaning; global production and work; environmental life-cycle performance; and intelligent or adaptive textiles. The review synthesizes peer-reviewed literature and authoritative institutional reports identified through targeted searches of publisher and public-agency databases. Global fibre production reached approximately 132 million tonnes in 2024, with polyester accounting for about 59% and cotton about 19%, illustrating the continuing dominance of high-volume fibre systems and the growing material throughput of apparel. Environmental burdens arise at different stages: feedstock and fibre cultivation or synthesis, wet processing, garment assembly, logistics, laundering and drying, microfibre release, and end-of-life loss. No fibre is universally superior across climate, water, land, toxicity, durability and recyclability indicators; therefore, fibre substitution alone is insufficient. The most robust cross-cutting strategy is to deliver more safe and valued service from fewer garments through fit, quality, durability, repairability, care optimization, reuse and credible recovery systems. Social performance is equally central because clothing supply chains combine economic opportunity with persistent risks related to wages, excessive hours, chemical exposure, building safety and unequal bargaining power. Smart textiles may expand clothing into continuous sensing, thermal management and human-machine interaction, but they also introduce new problems of power, privacy, wash durability, repair and electronic waste. The paper proposes a nested clothing-systems framework and a preferred intervention hierarchy—avoid, reduce, extend, reuse, recycle and controlled disposal. It concludes that the future of clothing should be evaluated not by novelty or unit price alone, but by inclusive human service per unit of material, energy, ecological pressure and labour risk.

Keywords: clothing; textiles; apparel; fashion; fibre; durability; circular economy; labour; microfibres; smart textiles; sustainability; human thermal comfort

Highlights

- Clothing is a nested body–garment–wardrobe–supply-chain–society–biosphere system.
- Global fibre output was approximately 132 million tonnes in 2024; polyester represented about 59%.
- No single fibre is best across all environmental, functional and social indicators.
- More wears, durable design, repair and high-quality reuse are generally more reliable than rapid replacement.

- Smart clothing requires simultaneous design for comfort, data governance, wash durability, repair and end-of-life recovery.

1. Introduction

Clothes are among the most intimate technologies created by humans. They remain in prolonged contact with the body, modify heat and moisture exchange, reduce exposure to mechanical and environmental hazards, enable occupational and ceremonial roles, and communicate identity. Unlike many manufactured objects, garments are simultaneously physiological interfaces, cultural signs, tradable commodities and material stocks. A satisfactory science of clothes must therefore cross textile engineering, physiology, chemistry, anthropology, design, economics, public health, labour studies, environmental assessment and digital technology.

The contemporary clothing system is characterized by unprecedented material scale and rapid product turnover. Textile Exchange estimated that global fibre production reached about 132 million tonnes in 2024, more than twice the level at the beginning of the century; polyester accounted for 59% and cotton about 19% [1]. UNEP reports that the textile sector contributes a material share of global greenhouse-gas emissions, water demand, chemical use and microplastic pollution, although exact estimates vary with system boundaries and methods [2-4]. The magnitude of these burdens makes it misleading to reduce the problem to consumer preference or recycling alone.

At the same time, clothes cannot be treated only as an environmental liability. They are essential for dignity, safety, thermal adaptation, public participation and livelihoods. The sector supports employment across agriculture, petrochemicals, fibre and fabric production, dyeing, logistics, design, retail, repair and reuse. Yet it also contains severe asymmetries of power and risk. International guidance continues to emphasize occupational safety, working hours, wages, freedom of association, supply-chain due diligence and the need for a just transition [5-7].

This paper develops an integrative account of clothing as human infrastructure. Its objectives are to: (i) define the major functions and levels of the clothing system; (ii) compare principal fibre and garment pathways without assuming a universally “green” material; (iii) identify environmental and social hotspots across the life cycle; (iv) assess the promise and limitations of circular and smart clothing; and (v) propose a research and policy agenda centred on useful service, durability, equity and verifiable outcomes.

2. Scope and Review Method

This manuscript uses an integrative review approach rather than a formal systematic-review protocol. Peer-reviewed reviews and representative empirical studies were combined with reports and standards from the United Nations Environment Programme (UNEP), International Labour Organization (ILO), Organisation for Economic Co-operation and Development (OECD), European Commission, European Environment Agency (EEA), Textile Exchange and the International Organization for Standardization (ISO). Targeted searches were conducted through publisher, PubMed and institutional web databases using combinations of the terms clothing, apparel, textiles, fibre production, thermal comfort, identity, fast fashion, life-cycle assessment, reuse, recycling, microfibres, garment labour, circular textiles, electronic textiles and smart clothing.

Sources were prioritized when they were peer reviewed, methodologically transparent, institutionally authoritative, widely cited, or recent enough to describe the current material and policy landscape. The synthesis is organized around causal mechanisms and decision points rather than a vote-counting comparison of heterogeneous studies. Quantitative figures are presented with their year and scope because clothing-impact estimates differ according to geography, product type, allocation rules, use assumptions and the inclusion or exclusion of footwear, household textiles, infrastructure and consumer travel.

The review has three limitations. First, it is broad and cannot replace product-specific life-cycle assessment, exposure testing or labour-rights investigation. Second, the evidence base is uneven: fibre production is relatively well quantified, whereas actual garment longevity, informal repair, post-export reuse and microfibre fate remain uncertain. Third, much published work focuses on Europe, North America and large export economies, while indigenous, informal and locally circular clothing systems are underrepresented.

Clothing as a nested human-infrastructure system

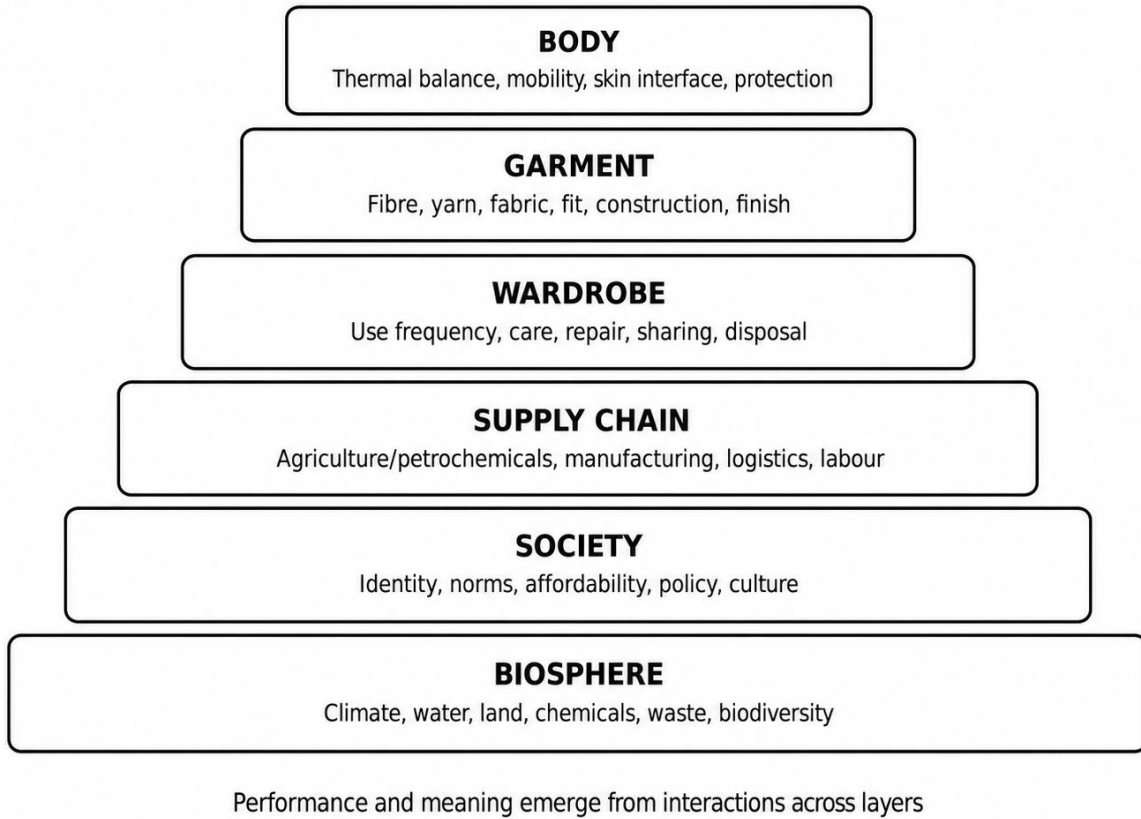


Figure 1. Clothing as a nested socio-technical system. The framework is an author synthesis: garment outcomes arise from interactions among body needs, product design, wardrobe practices, supply chains, social institutions and ecological limits.

3. Clothing as Human Infrastructure

3.1 Physiological protection and thermal regulation

The minimum function of clothing is to regulate exchange between the body and its environment. Garments alter conductive, convective, radiative and evaporative heat transfer; buffer solar load and wind; manage liquid sweat; and create barriers against abrasion, ultraviolet radiation, chemicals, pathogens, flame, cold or impact. Their effectiveness depends not only on fibre chemistry but also on yarn geometry, fabric porosity, thickness, layering, fit, body motion, moisture and environmental conditions.

Clothing insulation is often expressed in clo units. Under the conventional relation, one clo corresponds to a thermal resistance of $0.155 \text{ m}^2 \cdot \text{K} \cdot \text{W}^{-1}$, so clothing insulation may be represented as $I_{cl} = R_{cl}/0.155$, where R_{cl} is clothing thermal resistance in SI units [8]. The equation is useful but incomplete because real garments contain openings, pumping effects

and nonuniform air layers. Evaporative resistance can be equally important in warm or active conditions, especially where protective barriers inhibit sweat evaporation.

Thermal comfort is therefore a system outcome rather than a material label. A lightweight, porous garment may be effective in a hot-dry context but inadequate under high solar radiation or occupational exposure. A highly insulating system may protect in cold conditions while creating heat strain during work. Inclusive design must account for age, disability, pregnancy, body diversity, cultural dress, medication, metabolism and access to controlled indoor environments.

3.2 Mobility, fit, dignity and accessibility

Clothing changes how a body can move, sit, work, rest and be perceived. Fit is not a cosmetic afterthought: poor fit can create pressure, chafing, restricted range of motion, trip hazards, thermal discomfort and exclusion from school or employment. Standard sizing systems often fail because populations vary across stature, proportion, age and disability, while brand-specific grading creates inconsistency. Adaptive clothing can reduce dressing effort through alternative closures, seated-wear patterning, discreet access points and sensory-aware construction.

Dignity also has a material dimension. Garments must be affordable, socially acceptable, washable and repairable within the user's circumstances. An technically advanced item that requires unavailable detergents, charging systems or specialist repair may deliver less real service than a simpler garment supported by local knowledge and infrastructure.

3.3 Identity, communication and social order

Dress is a visible component of identity formation and person perception. Roach-Higgins and Eicher defined dress broadly as modifications and supplements to the body, allowing analysis of clothing together with hair, cosmetics, jewellery and other forms of bodily presentation [9]. Later experimental work confirms that clothing is rapidly integrated into judgments about people, even when observers are instructed to focus on other information [10].

Clothes can indicate occupation, age, religion, gender performance, political belonging, mourning, celebration, status and resistance. The same garment can carry different meanings across communities or historical periods. Consequently, universal prescriptions for “sustainable style” risk cultural simplification. Effective interventions should protect cultural continuity and user agency while challenging coercive norms, discrimination and wasteful status competition.

Table 1. Principal functions of clothing and corresponding evaluation criteria.

Function	Examples	Evaluation criteria	Typical failure modes
Protection	Cold, heat, UV, flame, chemicals, impact, pathogens	Barrier performance; thermal and evaporative resistance; hazard-specific standards	Insufficient protection; heat stress; barrier degradation; false sense of safety
Comfort and mobility	Daily wear, sport, sleep, work	Fit; pressure; stretch; mass; moisture transport; tactile response	Chafing; restricted motion; overheating; sensory irritation
Identity and culture	Religious dress, ceremony, uniforms, subculture	Meaning; autonomy; cultural legitimacy; non-discrimination	Stigmatization; appropriation; compulsory conformity
Economic service	Employment, school participation, affordability	Cost per wear; repair access; durability; availability	Low quality; debt; exclusion; premature disposal
Information and intelligence	Sensing, heating, communication, adaptive response	Accuracy; privacy; power; washability; repairability	Data misuse; battery failure; e-waste; discomfort
Environmental stewardship	Resource use and recovery	Life-cycle impacts; longevity; reuse; recyclability; toxicity	Burden shifting; greenwashing; downcycling; leakage

4. Materials and Manufacturing

4.1 Fibre classes and trade-offs

Fibres used in clothing can be grouped into natural fibres, regenerated cellulosic fibres and synthetic fibres, with additional niches for mineral, metallic and emerging biofabricated materials. Their environmental profiles are multidimensional. Cotton is renewable and familiar but can involve irrigation, pesticide use and land competition; wool offers insulation, moisture buffering and potential longevity but is associated with livestock

impacts and specific animal-welfare questions; regenerated cellulose can deliver soft, moisture-responsive fabrics while depending on forestry practice and solvent recovery; polyester is durable, inexpensive and quick drying but is primarily fossil based and can shed persistent microplastic fibres.

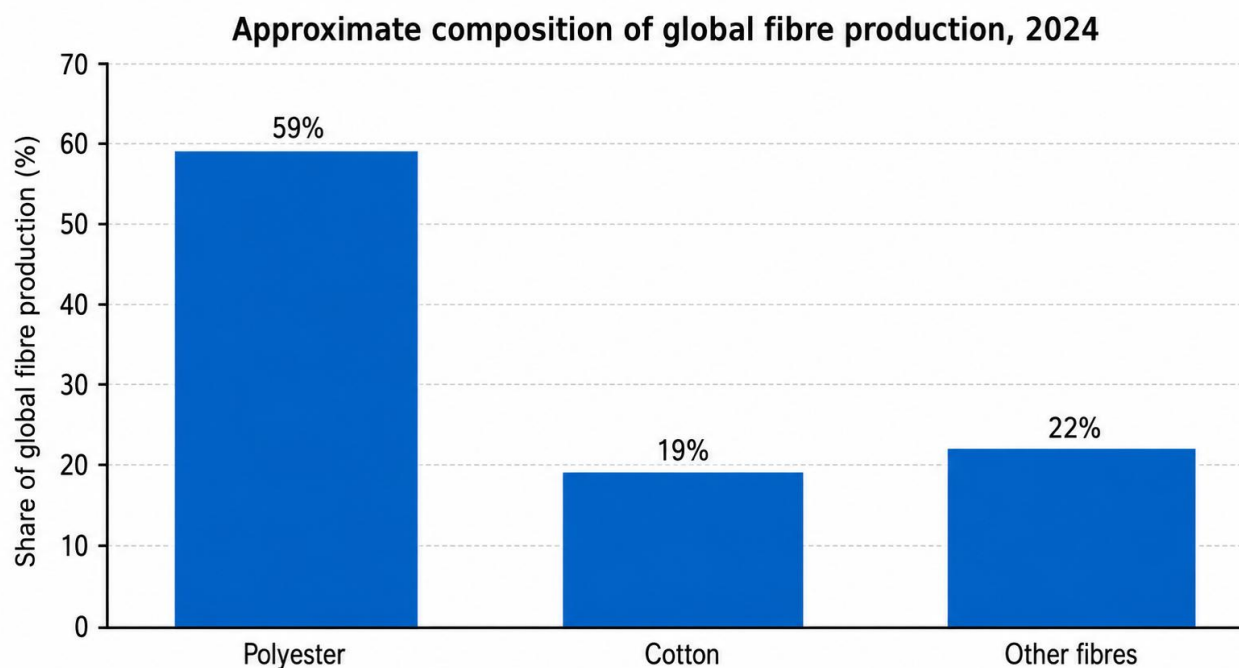
The present market is dominated by synthetics. In 2024, polyester represented approximately 59% of world fibre production, while recycled polyester reached only about 12% of polyester output and remained largely dependent on non-textile feedstocks such as bottles [1]. This creates a dual challenge: reducing virgin fossil input and building fibre-to-fibre systems that preserve material quality without increasing total throughput.

Blends are used because functions are complementary: cotton-polyester can combine hand feel with strength and dimensional stability; elastane can improve stretch and recovery; coatings can add water resistance or flame performance. However, blended and multilayer garments are difficult to separate and recycle. Design decisions that improve short-term performance may therefore create end-of-life lock-in.

Table 2. Comparison of major clothing-fibre families. Assessments are conditional and should not be interpreted as universal rankings.

Fibre family	Functional strengths	Key sustainability concerns	Design priorities
Cotton	Breathable; familiar; washable; versatile	Water and agrochemical burden can be high; quality losses in mechanical recycling	Verified agricultural practice; durable yarns; lower-loss cutting; fibre-to-fibre routes
Wool and animal fibres	Insulation; moisture buffering; odour resistance; potential long life	Livestock emissions, land use, animal welfare, moth damage	Traceability; long use; repair; responsible husbandry; careful laundering
Bast fibres (linen, hemp)	Strength; low mass; distinctive moisture behaviour	Processing intensity, variability, limited scale in some markets	Regional processing; quality consistency; design for longevity
Regenerated cellulose	Softness; drape; moisture response	Forestry sourcing, chemical recovery, wet-processing impacts	Certified feedstock; high solvent recovery; responsible chemistry

Fibre family	Functional strengths	Key sustainability concerns	Design priorities
Polyester	Strength; dimensional stability; quick drying; low cost	Fossil feedstock; microfibre release; low textile-to-textile recycling share	Virgin reduction; low-shed construction; mono-material design; high-quality recycling
Polyamide/nylon	Abrasion resistance; elasticity; technical performance	Fossil feedstock; energy intensity; microfibre release	Use where performance extends life; repair; recycled feedstock with verified benefit
Elastane and functional additives	Stretch, fit, compression, specialised protection	Small fractions can obstruct recycling; chemical complexity	Minimum effective content; detachable components; transparent material passports



Global fibre production: approximately 132 million tonnes.
Polyester share includes predominantly fossil-based production; cotton was the second-largest fibre group.
Shares are approximate and rounded.

Figure 2. Approximate global fibre-production shares in 2024. Data: Textile Exchange, Materials Market Report 2025 [1]. “Other fibres” combines all fibre groups other than polyester and cotton.

4.2 From yarn to garment

Fibre properties are transformed by spinning, fabric formation and finishing. Staple length, yarn twist, filament texturing, knit or weave architecture, stitch density and surface finishing influence strength, pilling, abrasion, drape, air permeability and shedding. The garment stage adds seams, trims, interlinings, prints, labels, hardware and pattern geometry. Quality failures frequently originate at interfaces—such as seam slippage, incompatible shrinkage or coating delamination—rather than in the nominal fibre itself.

Wet processing is a major environmental and occupational hotspot because preparation, dyeing, printing and finishing can require heat, water, salts, auxiliaries and complex chemistry. Cleaner production includes process control, low-liquor systems, heat recovery, wastewater treatment, restricted-substance management, safer substitution and verification of discharge rather than reliance on product-level claims alone [2,3].

Cut-and-sew efficiency also matters. Pattern layout determines pre-consumer waste, while workmanship determines service life. Zero-waste and whole-garment methods can reduce cutting loss, but their net benefit depends on energy, equipment, fit, product acceptance and whether avoided waste leads to lower material throughput rather than simply lower unit cost.

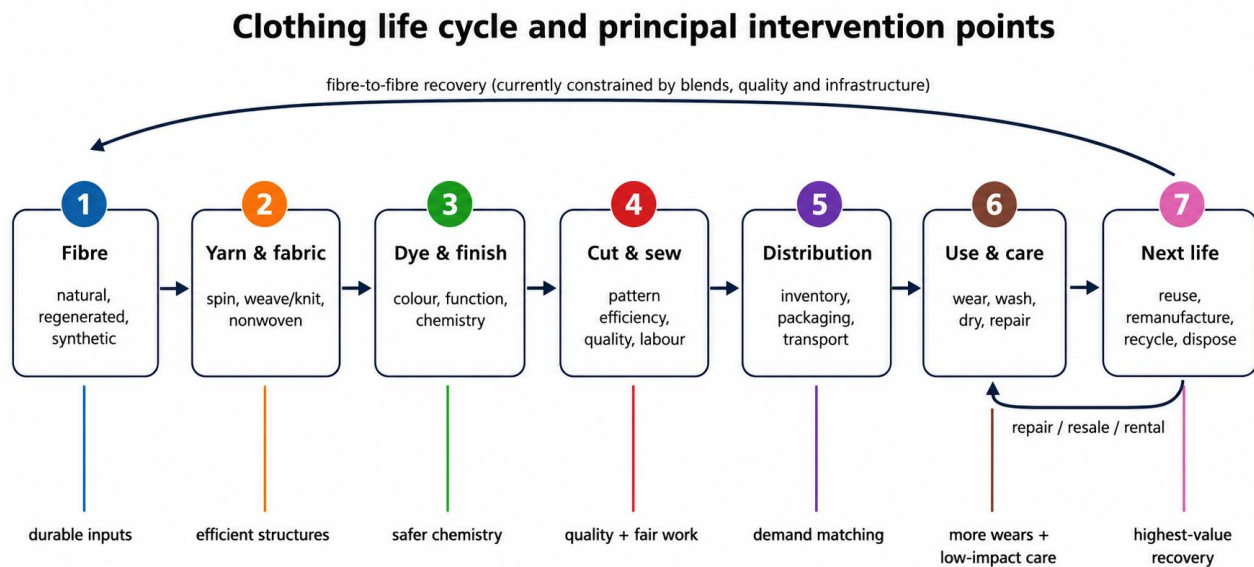


Figure 3. Clothing life cycle and intervention points. The diagram emphasizes that circularity is not an end-of-life activity alone; decisions made during fibre

selection, fabric construction, finishing and garment design determine later durability, care and recovery options.

5. Performance, Comfort and Health

5.1 Skin interface, moisture and hygiene

Garments form a variable microclimate next to the skin. Moisture can be transported as vapour, absorbed into fibres, wicked through capillaries or retained in fabric pores. The preferred behaviour depends on activity and climate. Rapid wicking may help spread sweat but can also increase evaporative cooling; absorbent fibres may buffer transient moisture but dry slowly; hydrophobic fabrics may remain light while accumulating odour-active soils.

Claims such as “antibacterial,” “cooling” or “moisture managing” should be interpreted through standardized testing and realistic use. Finishes may lose effectiveness after laundering, alter skin exposure or complicate recycling. For everyday clothing, cleanliness, drying, fit and opportunity for rotation may contribute more to hygiene than durable biocidal treatment. Medical and protective garments require hazard-specific validation rather than extrapolation from consumer tests.

5.2 Chemical exposure and occupational health

Chemical risk occurs at multiple points: workers may encounter dusts, dyes, solvents, resins, adhesives and finishing agents; communities may be exposed through air and water releases; users may encounter residual substances or degradation products. Risk depends on hazard, dose, route and duration, so the presence of a chemical does not by itself establish harm. Nevertheless, the ILO code of practice emphasizes elimination and substitution where feasible, exposure control, worker participation, training, medical surveillance and provision of effective protective systems without shifting costs to workers [6].

Health analysis must also avoid burden transfer. A coating that reduces laundering may save water but introduce persistent chemistry; a reusable protective garment may reduce waste but require energy-intensive decontamination; a lightweight garment may lower transport impacts yet wear out sooner. Decisions should compare complete service systems under realistic conditions.

5.3 Clothing poverty and public health

Access to adequate clothing is a social determinant of participation and safety. Insufficient seasonal clothing can worsen cold or heat exposure; lack of uniforms or socially acceptable

garments can restrict education and employment; poorly fitted footwear and clothing can impair mobility; and laundry insecurity can contribute to stigma. Clothing policy is therefore connected to housing, energy, water, sanitation, public health and social protection.

Donation programmes should be demand-led and quality controlled. Exporting unsorted low-value garments can shift sorting and disposal burdens to receiving regions. High-quality redistribution, local repair, fibre identification and transparent downstream accountability are preferable to treating collection volume as proof of circularity.

6. Environmental Burdens Across the Life Cycle

6.1 Scale, climate, water and land

The environmental price of clothing arises from scale multiplied by per-unit impact. A widely cited review estimated more than 92 million tonnes of textile waste and approximately 79 trillion litres of water consumption annually for the fashion sector under its selected boundaries [11]. Such global estimates are useful indicators of magnitude but should not be applied to individual garments without product-specific data.

Climate impacts differ by fibre and geography. Synthetic fibres depend heavily on fossil energy and feedstocks; cotton impacts are sensitive to irrigation and yield; animal fibres can be dominated by enteric emissions and land-use assumptions; wet processing and consumer drying can be energy intensive. Life-cycle assessment therefore requires transparent functional units. Comparing one kilogram of fibre may produce a different conclusion from comparing one garment, one year of service or one hundred wears.

A service-based metric is conceptually preferable. For an impact category E, impact per wear can be represented as $E_{\text{wear}} = E_{\text{lifecycle}}/N_{\text{wear}}$, where N_{wear} is the number of effective wears before retirement. The equation is simple, but it captures why quality, fit, emotional attachment, repair and care are environmental variables. It also shows why unused durable garments do not automatically perform well: service must actually be delivered.

6.2 Microfibres and diffuse pollution

Textile fibres are released during manufacturing, wearing, laundering, drying and disposal. Studies have demonstrated that domestic washing of synthetic fabrics can release large numbers of fibres, with results varying by fabric type, construction, age, wash conditions and measurement method [12-15]. The environmental behaviour of fibres also depends on polymer, additives, size, shape and receiving system.

End-of-pipe capture can reduce emissions, but source-oriented design is essential. Tighter, stable structures; controlled surface hairiness; durable finishes; edge construction; prewashing controls; quality standards; and reduced abrasion can lower shedding. A 2024 experimental study found that, among tested polyester constructions, single-jersey knit released more than three times the microplastic mass of twill woven fabric, illustrating the importance of structure rather than fibre label alone [16].

Natural fibres also fragment, and their additives and degradation rates vary. Research and policy should distinguish persistent plastic microfibres from broader textile-fibre pollution while avoiding the assumption that all natural fragments are harmless or that all recycled synthetics solve shedding.

6.3 Waste, reuse and uncertain destinations

Waste statistics reveal both high throughput and limited recovery. In the European Union, apparent consumption of clothing, footwear and household textiles reached about 19 kg per person in 2022, while textile waste generation was about 16 kg per person [17,18]. These are regional figures, not global averages, but they demonstrate the challenge of matching collection systems with sorting, reuse and recycling capacity.

Reuse and recycling usually provide environmental benefit when they displace new production and when collection, transport and processing do not outweigh avoided burdens. A major review found that reuse typically offers greater benefits than recycling, but outcomes depend on replacement rates, transport, energy and product quality [19]. Exported used clothing has uncertain fates when receiving markets are saturated or waste infrastructure is limited [20].

Technical recycling includes mechanical opening, dissolution, depolymerization and hybrid processes. Mechanical recycling is established but often shortens fibres and lowers quality. Chemical routes can recover higher-quality feedstock for selected polymers or cellulose but require purity, energy, chemicals and infrastructure. The strategic objective should be high-value closed-loop recovery after a long useful life—not recycling as permission for accelerated consumption.

Table 3. Life-cycle hotspots, evidence needs and preferred interventions.

Stage	Typical burdens	Evidence needed	Priority interventions
Raw materials	Land, water, fertilizer, livestock emissions, fossil extraction	Region-specific inventory; biodiversity and social data	Lower virgin demand; verified production; recycled or regenerative inputs where beneficial

Stage	Typical burdens	Evidence needed	Priority interventions
Fibre/fabric production	Energy, dust, solvents, waste, worker exposure	Process energy; chemical recovery; exposure monitoring	Efficiency, renewable energy, capture and substitution, worker protection
Dyeing/finishing	Heat, water, salts, hazardous releases	Facility discharge and chemical-management data	Safer chemistry, low-liquor processes, treatment, heat recovery
Garment assembly	Cutting loss, low wages, excessive hours, quality defects	Purchasing practices; wages; hours; defects; safety	Fair purchasing, quality systems, worker voice, safer buildings
Distribution/retail	Overproduction, returns, packaging, transport	Inventory destruction, return rates, mode and distance	Demand matching, slower assortment cycles, ban avoidable destruction
Use/care	Washing, drying, ironing, microfibre release	Actual wear and care behaviour	More wears, lower-temperature wash, line drying where suitable, repair
End of life	Landfill, incineration, export leakage, downcycling	Destination tracking; displacement; recovery yield	Local reuse, verified exports, design for sorting, fibre-to-fibre recovery

7. Consumption, Durability and Circularity

7.1 The durability paradox

Durability has physical, functional, emotional and social dimensions. Physical durability is resistance to rupture, abrasion, pilling, colour loss and dimensional change. Functional durability is continued ability to meet the user's needs. Emotional durability concerns attachment and willingness to keep wearing. Social durability reflects whether the garment remains acceptable within work, school or community expectations.

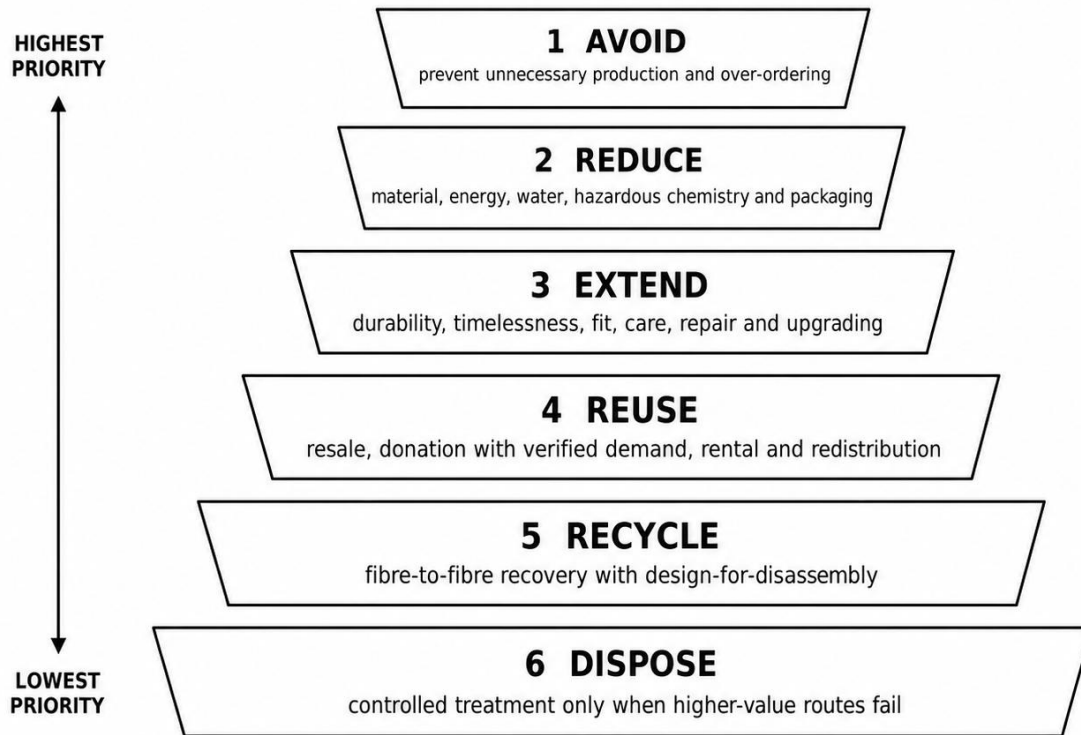
A garment can be physically robust but functionally obsolete because it no longer fits, cannot be repaired, depends on unavailable electronics or was designed around a short-lived trend. Conversely, a well-loved garment with visible repairs may deliver high service. Evidence across fibre types indicates that increasing the number of wears is consistently beneficial for environmental performance [21]. The implication is that durability policy must connect measurable quality to sizing, design, care information, repair access and demand reduction.

7.2 Circular business models

Resale, rental, repair subscriptions, take-back systems and product-service models can increase utilization, but none is automatically sustainable. Rental can add transport and cleaning; take-back can become a collection claim without high-value destinations; resale can coexist with continued high primary consumption; repair services may be priced beyond reach. Evaluation should measure additional wears, displacement of new production, logistics, cleaning, labour conditions and final material fate.

Consumer research repeatedly identifies an attitude-behaviour gap: people may endorse sustainability while prioritizing price, novelty, identity and convenience [22]. Information is necessary but insufficient. Product quality, market incentives, default options, repair infrastructure, marketing pressure and social norms shape behaviour. Responsibility should therefore be distributed among producers, retailers, governments, financiers, platforms and consumers rather than placed on the individual buyer alone.

Preferred intervention hierarchy for clothing systems



Priority decreases downward; technical recycling should not justify avoidable throughput.

Figure 4. Preferred intervention hierarchy for clothing. Avoidance and service-life extension usually preserve more embedded value than material recovery. Recycling remains necessary but should follow prevention, reduction, repair and reuse.

Table 4. Circular strategies and conditions for genuine benefit.

Strategy	Potential benefit	Common rebound or failure	Minimum evidence for credibility
Durable design	More service per garment	Overengineering; low actual use; higher sales volume	Measured quality plus observed or plausible additional wears
Repair	Extends life and preserves value	Parts unavailable; cost exceeds replacement; design blocks access	Repairability, spare parts, instructions, skilled labour and uptake
Resale/reuse	Displaces some new production	Low displacement; export dumping; rapid turnover continues	Destination tracking, quality grading and displacement analysis
Rental/sharing	Higher utilization of infrequent-use items	Transport, cleaning, damage and novelty rebound	Full service-system LCA and utilization data
Recycling	Reduces virgin feedstock and waste	Downcycling; contamination; high energy; low yield	Mass balance, yield, quality, energy source and actual secondary use
Digital product passport	Improves identification and accountability	Incomplete data; proprietary lock-in; privacy and access barriers	Interoperability, verification, long-term access and governance

8. Labour, Equity and Governance

8.1 Work and value distribution

Clothing supply chains distribute tasks and risks unevenly. Brands and retailers often control design, pricing and purchasing schedules, while suppliers operate under intense cost and lead-time pressure. These commercial practices can contribute to excessive overtime, unauthorized subcontracting, wage violations and underinvestment in safety. The ILO identifies excessive working hours and occupational hazards as continuing concerns, while

OECD guidance places responsibility on enterprises to identify, prevent, mitigate and account for adverse impacts across their supply chains [5-7].

Women constitute a large share of garment employment in many production regions and may face occupational segregation, wage gaps, harassment and unequal care burdens. A just transition to automation, circularity and lower material throughput must not treat workers as disposable. Skills development, social protection, worker representation, predictable purchasing, living wages, occupational health and formalization of repair and waste work are part of sustainability rather than separate social add-ons.

8.2 Policy transition

Policy is shifting from voluntary claims toward product requirements, producer responsibility and due diligence. The European Union's textiles strategy links durability, repairability, recycled content, hazardous-substance control, digital product information and extended producer responsibility [23]. Separate collection of used textiles became mandatory for EU Member States from 2025, and 2026 measures under the Ecodesign for Sustainable Products Regulation began addressing destruction of unsold apparel and footwear [24,25].

These measures are important experiments, but regulation should avoid metric fixation. A recycled-content target can increase bottle-to-textile flows without creating textile-to-textile circularity; a durability standard can encourage heavier products regardless of actual service; a product passport can become a compliance label without verified data. Policy portfolios should combine absolute material-demand indicators, quality and repair requirements, chemical safety, labour due diligence, collection and sorting infrastructure, anti-greenwashing rules and public procurement.

9. Smart and Adaptive Clothing

9.1 Capabilities

Smart textiles integrate sensing, actuation, communication, energy management or computation into fibres, yarns, fabrics or garments. Applications include electrocardiography, respiration and motion monitoring; pressure mapping; rehabilitation; worker safety; sports feedback; personal heating and cooling; displays; haptics; and environmental sensing. Textile platforms offer flexibility, breathability and large-area body contact, which can be advantageous over rigid devices [26-30].

Smart-clothing systems can be described as five coupled modules: textile interface, sensor or actuator, interconnection, power, and data interpretation. Failure in any module can

invalidate the whole garment. A highly accurate sensor is not useful if electrodes lose contact during movement, interconnects break in washing, batteries create stiffness, algorithms are biased, or the user cannot repair or understand the device.

9.2 Thermal and health applications

Personal thermal-management textiles may reflect or emit infrared radiation, vary insulation, move moisture, store heat through phase-change materials or generate heat electrically. Such systems could reduce dependence on conditioning entire buildings when local control is acceptable, but benefits depend on climate, behaviour, electricity source, safety and rebound [31].

Health-monitoring garments may provide unobtrusive longitudinal data, yet clinical validity cannot be inferred from laboratory signal quality. Validation must address diverse bodies, motion artefact, laundering, calibration drift, missing data and outcomes that matter to patients. Privacy is fundamental because clothing may collect intimate physiological, location and behavioural information.

9.3 Circular design for e-textiles

Electronic textiles combine components with very different lifetimes and recovery pathways. Conductive inks, metal-coated yarns, batteries, adhesives, encapsulants and conventional fibres can make disassembly difficult. A circular smart garment should use detachable high-value modules, standardized connectors, replaceable power, documented materials, software support, secure deletion, repair pathways and end-of-life routing.

The design objective is not maximum electronics per garment, but the minimum technological complexity needed to deliver verified human benefit. Passive textile engineering should remain the baseline; electronics are justified when they add service that cannot be achieved reliably through simpler construction.

Table 5. Research priorities for next-generation clothing systems.

Priority	Core question	Recommended methods	Success indicator
Service-based assessment	How much valued service is delivered per unit of impact?	Longitudinal wardrobe studies; wear counters; functional-unit harmonization	Verified increase in useful wears and user benefit

Priority	Core question	Recommended methods	Success indicator
Inclusive fit and comfort	Who is excluded by current sizing and testing?	3D anthropometry; participatory design; field trials across bodies and climates	Lower discomfort, returns and exclusion across diverse users
Low-shed textiles	Which structures and processes minimize fibre release?	Standardized manufacturing, wear and wash tests; fate studies	Lower release without sacrificing longevity or safety
Textile-to-textile recycling	How can mixed garments be separated at high yield and quality?	Design-for-disassembly, sorting, chemical/mechanical process trials, mass balance	High secondary-fibre quality and verified virgin displacement
Fair circular work	Who gains and who bears risk in repair, sorting and recycling?	Labour mapping, wage and exposure studies, worker-led governance	Decent work, safer exposure and equitable value distribution
Smart-garment governance	Can sensing garments remain accurate, private, repairable and recyclable?	Clinical validation, cybersecurity testing, wash cycling, modular design	Verified benefit with low failure, protected data and recoverable modules

10. An Integrative Evaluation Framework

The evidence supports a shift from product-centred to service-centred evaluation. A garment should be assessed as a proposition to deliver protection, comfort, identity, participation or information over time. The proposed framework contains six questions:

1. Human service: What need is being met, for whom, under which climate, activity and cultural conditions?
2. Functional adequacy: Does the garment meet verified requirements for fit, comfort, protection and durability?
3. Material sufficiency: Is the least resource-intensive configuration used without compromising safety or service life?

4. Social legitimacy: Are workers, users and affected communities protected, represented and fairly compensated?
5. Life-cycle performance: Where are the dominant climate, water, land, toxicity, microfibre and waste burdens?
6. Next-life readiness: Can the garment be maintained, repaired, reused, disassembled and recovered through real infrastructure?

For decision making, the framework can be operationalized as a dashboard rather than a single score. Aggregated scores obscure trade-offs and value judgments; a garment with excellent carbon performance but unacceptable labour conditions should not be considered sustainable. Minimum thresholds for safety, human rights and chemical hazard should be treated as non-compensable constraints, after which environmental and service trade-offs can be optimized.

A useful reporting unit is “verified service per garment,” expressed through wears, hours of protection, occupational shifts, thermal-comfort hours or clinical-monitoring days. This encourages alignment between engineering tests and lived use. It also makes overproduction visible: products that are never used deliver zero service regardless of efficient manufacture.

11. Research and Policy Agenda

First, researchers should build comparable datasets on actual use. Purchase records are not wear records, and donation is not confirmed reuse. Privacy-preserving wardrobe panels, repair histories and product-level identifiers could improve estimates of service life and displacement.

Second, standards should connect laboratory durability to failure modes observed in households and workplaces. Test methods should cover seams, colour, dimensional stability, pilling, coatings, trims and repair, while recognizing that acceptable appearance is culturally variable.

Third, chemical and microfibre policy should move upstream. Facility controls, safer substitution, construction standards and shedding tests are more reliable than placing full responsibility on laundry behaviour.

Fourth, circular infrastructure requires economic coordination. Collection without sorting markets creates stockpiles; recycling plants without compatible design lack feedstock; repair services without quality products lack viable demand. Extended producer responsibility should fund transparent systems while avoiding incentives for low-value collection volume.

Fifth, trade and labour governance must address purchasing practices. Lead times, price negotiation, order changes and payment terms influence whether suppliers can provide safe work and quality. Due diligence should include remedy, worker participation and consequences for repeated harm.

Sixth, smart clothing research should adopt “safe, private, repairable and separable by design” as a default. Clinical or safety claims require appropriate validation, and software support periods should match expected garment and component life.

Finally, policy should monitor absolute material throughput as well as efficiency. Efficiency gains can be overwhelmed by increasing volume. The transition requires fewer avoidable garments, higher utilization, better essential access and protected livelihoods—not merely a different material mix.

12. Limitations

This review integrates heterogeneous evidence and does not provide a meta-analysis. Global industry statistics are periodically revised, and the boundaries of “fashion,” “textiles,” “clothing” and “apparel” differ among sources. Environmental comparisons are highly sensitive to geography and functional unit. Social conditions vary by facility and cannot be inferred from country averages. Smart-textile evidence often comes from prototypes or short laboratory trials and may overstate readiness for long-term use.

The paper also necessarily simplifies cultural diversity. Clothing practices that appear inefficient under industrial accounting may support identity, care, local livelihoods and material knowledge. Conversely, claims of tradition should not shield coercion, discrimination or hazardous work. Future work should use participatory and regionally grounded methods.

13. Conclusion

Clothes are not trivial consumer goods. They are body-scale infrastructure that supports survival, mobility, dignity, identity and work while linking households to global material, labour and ecological systems. The current system delivers these benefits through a volume-intensive model dominated by synthetic fibre growth, fragmented accountability and weak end-of-life recovery.

The review finds no universally sustainable fibre and no single technological fix. The strongest strategy is a coordinated portfolio: prevent avoidable production; design for adequate protection and inclusive fit; use safer and more efficient processes; secure decent work; maximize useful life through quality, care and repair; enable high-quality reuse; and

build fibre-to-fibre recovery for the remaining material. Smart textiles should be judged by verified human benefit and designed for privacy, repair and separation.

The central research and policy question is therefore not “What is the greenest garment?” but “How can clothing systems deliver equitable human service with the least material throughput, ecological damage and labour risk?” Answering that question requires engineering, cultural understanding, public policy and worker and user participation to be treated as one field of inquiry.

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Appendix A. Clothing-System Assessment Checklist

- Define the user, climate, activity, cultural setting and required service.
- Specify protective, comfort, mobility, appearance and care requirements.

- Identify all fibres, coatings, trims, electronics, adhesives and packaging.
- Test foreseeable failure modes, including seams, pilling, colour, shrinkage and component fatigue.
- Estimate expected wears or service hours under realistic use and care.
- Assess facility-level energy, water, chemistry, emissions and worker protection.
- Examine purchasing practices, wages, hours, worker voice and remedy mechanisms.
- Provide clear care, repair, parts and warranty information.
- Design for identification, disassembly and the highest-value available next-life route.
- Report uncertainties and avoid unsupported “sustainable,” “eco-friendly” or “circular” claims.